

Experiment 'p_6_2_b_shift_right_seq' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_b_shift_right_seq

Natural language statement: Write a method to shift all elements of an array by one to the right and move the last element into the first position. For example, [1, 4, 9, 16, 25] would be transformed into [25, 1, 4, 9, 16].

Method signature: p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `result[0] == inputs[|inputs| - 1]`
- forall `i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]`

Requires

- `|inputs| >= 1`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to shift all elements of an array by one to the right and move the last
→ element into the first position. For example, [1, 4, 9, 16, 25] would be
→ transformed into [25, 1, 4, 9, 16].

The signature should be:

p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)

The method should respect the following contract:

requires |inputs| >= 1, ensures |result| == |inputs|, ensures result[0] == inputs[|inputs|
→ | - 1|], ensures forall i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0Coultx7ZpJFxnbySe1yLAJa3Iu

Model: gpt-5.5-2026-04-23

Created at: 1780571234

Finish reason: stop

Usage: 454 tokens in, and 578 tokens out

```
{  
  result := seq(|inputs|, i requires 0 <= i < |inputs| =>  
    if i == 0 then inputs[|inputs| - 1] else inputs[i - 1]);  
}
```

Final Program

```
method p_6_2_b_shift_right_seq(inputs: seq<int>) returns (result: seq<int>)  
  requires |inputs| >= 1  
  ensures |result| == |inputs|  
  ensures result[0] == inputs[|inputs| - 1]  
  ensures forall i :: 1 <= i < |inputs| ==> result[i] == inputs[i - 1]  
{  
  result := seq(|inputs|, i requires 0 <= i < |inputs| =>  
    if i == 0 then inputs[|inputs| - 1] else inputs[i - 1]);  
}
```

Total Token Usage

Input tokens: 454

Output tokens: 578

Reasoning tokens: 512

Sum of ‘total tokens’: 1032

Experiment Timings

Overall Experiment started at 1780571234479, ended at 1780571271197, lasting 36718ms (36.72 seconds)

Iteration #1 started at 1780571234484, ended at 1780571271197, lasting 36713ms (36.71 seconds)